

# Design Patterns for Car Rental Management System

## 1. Singleton Pattern

### Purpose

Ensures that only one instance of a class exists and provides a global access point.

### Why It Fits the Project

- Authentication system
- Database connection
- Reservation management
- Pricing management

### Related Use Cases

UC1 – Register Account, UC18 – Employee Login, UC30 – Administrator Login, UC4 – Create Reservation

### Justification

The Singleton pattern is used to ensure centralized control of important services such as authentication and database operations. It prevents multiple conflicting instances and improves consistency in the system.

### Suggested Classes

DatabaseConnection, AuthenticationManager, ReservationManager

## 2. Factory Method Pattern

### Purpose

Creates objects without exposing the exact object creation logic.

### Why It Fits the Project

- Payment methods
- Vehicle categories
- Different user roles

### Related Use Cases

UC13 – Select Payment Method, UC14 – Process Payment

### Justification

The Factory Method pattern allows the system to dynamically create different payment methods and other objects without tightly coupling the code to specific classes.

### Suggested Classes

PaymentFactory, PaymentMethod, CreditCardPayment, PayPalPayment, CashPayment

## 3. Observer Pattern

### Purpose

Allows objects to receive automatic updates when another object's state changes.

### Why It Fits the Project

- Reservation notifications
- Payment confirmations
- Vehicle availability updates

## Related Use Cases

UC20 – Confirm Reservation, UC21 – Generate Rental Agreement, UC23 – Update Vehicle Availability

## Justification

The Observer pattern is used for notifications and real-time updates between system components and users whenever important changes happen.

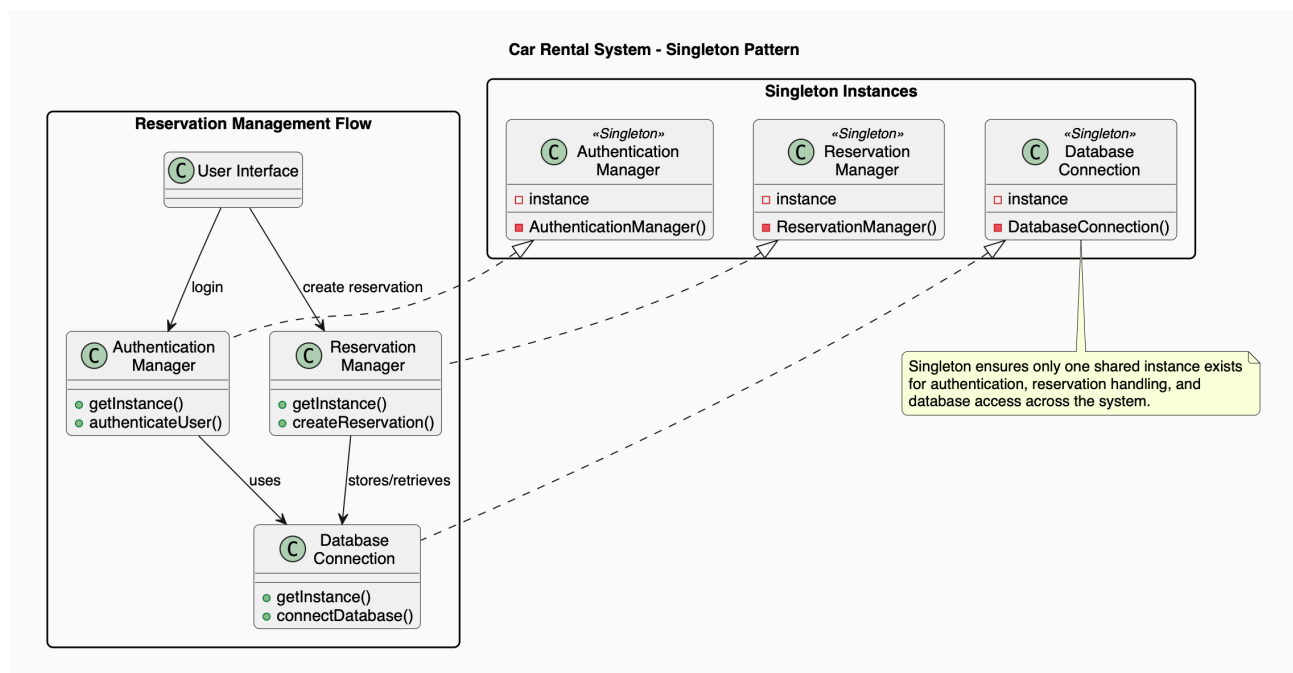
## Suggested Classes

Subject, Reservation, Observer, CustomerNotification, EmployeeNotification

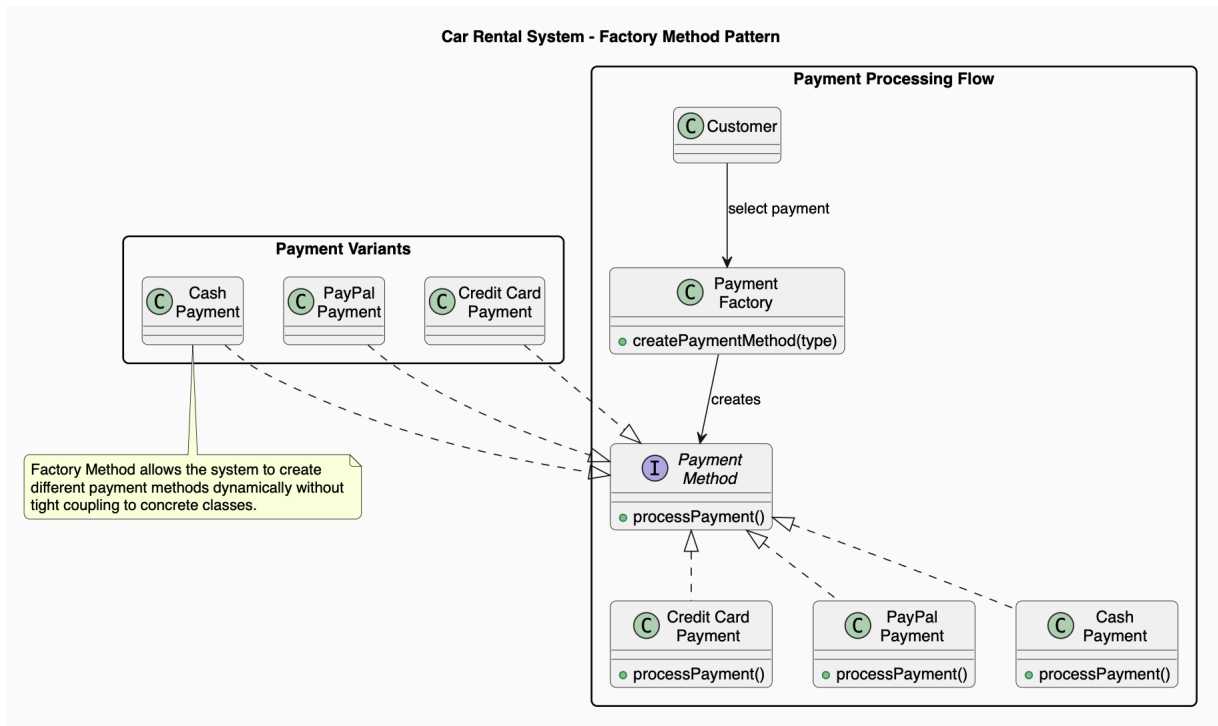
## Conclusion

The selected design patterns improve flexibility, maintainability, and organization of the Car Rental Management System. These patterns are directly connected to the project requirements and system use cases.

## -Singleton Pattern (Class Diagram)



## -Factory Method Pattern (Class Diagram)



## -Observer Pattern (Class Diagram)

